

Engineering Fairness: A Framework for Mitigating Algorithmic Bias in ML-Based Applicant Tracking Systems

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Abstract—ML-based Applicant Tracking Systems (ATS) are now a fixture in modern hiring, yet there is a troubling irony at their core: the very historical data they learn from often encodes the biases of the people who came before them. The result is a system that does not discriminate by design—it discriminates by inheritance. In this paper, we build an auditing framework that can catch this inherited bias and correct it. We anchor our fairness criterion in the EEOC “four-fifths rule” and apply a re-weighting technique to rebalance the training data before the model ever sees it. Testing on a dataset of 10,847 job applications, our Fair-RF model improved the Disparate Impact (DI) score from 0.62 to 0.84—crossing the legal compliance threshold—while only sacrificing 1.7% in accuracy (from 92.1% to 90.4%). The takeaway is straightforward: making a hiring system fairer does not have to mean making it worse at its job.

Index Terms—Algorithmic Bias, Applicant Tracking Systems, Demographic Parity, Disparate Impact, Equal Opportunity AI, Fair Machine Learning, Random Forest, Re-Weighting.

I. INTRODUCTION

There is something quietly troubling about the way AI-driven hiring tools work today. Most companies train these systems on their own historical records—years of decisions about who got hired and who did not. But those decisions did not happen in a vacuum. They were made by people, in organisations, shaped by the biases and assumptions that were simply part of the culture at the time. When a machine learning model learns from that data, it learns those patterns too, and carries them forward. This is sometimes called the “black box” problem: the system influences people’s careers without anyone being able to clearly explain why [1] [2].

What makes this especially tricky is that modern ATS platforms do not need to ask about someone’s gender or ethnicity to discriminate. They learn to use proxy variables instead—things like resume gaps, the prestige tier of a university, or even a postal code that happens to correlate with a particular demographic group. Barocas and Selbst (2016) described this as “disparate impact through data”: the discrimination is baked into the features, not the rules [2]. Figure 1 makes this concrete: two candidates with genuinely identical qualifications receive dramatically different ATS scores, for no reason the system would ever articulate.

Alex Johnson (Majority Group)	Priya Sharma (Minority – Same Qualif.)
M.Sc. CS – State Univ. Sr. Data Sci., TechCorp (4y) Resume Gap: 8 months ATS: 91 SELECTED	M.Sc. CS – State Univ. Sr. Data Sci., TechCorp (4y) Resume Gap: 8 months ATS: 58 REJECTED

Fig. 1. Proxy Variable Discrimination. Identical qualifications, divergent ATS scores. Candidate B is penalised by features acting as hidden demographic surrogates.

The legal stakes are real, and they fall on employers. The EEOC holds organisations accountable for discriminatory outcomes regardless of whether the bias was intentional [4]. Amazon’s experience is the clearest warning sign: the company quietly ran an AI hiring tool for years that consistently down-scored female applicants—despite gender never being an explicit input—before eventually shutting it down [5]. This paper proposes an auditable framework designed to catch and correct exactly this kind of problem before it does harm.

The contributions of this work are threefold: (i) an iterative bias-auditing loop grounded in the EEOC four-fifths rule, (ii) a pre-processing correction layer using re-weighting that is classifier-agnostic, and (iii) empirical validation on a real-world hiring dataset showing that legal compliance and predictive accuracy are not mutually exclusive.

II. RELATED WORK

Hiring technology has moved through a few recognisable phases. Early ATS systems were essentially keyword scanners—they looked for specific terms and passed or failed candidates based on pattern matches. Simple, and at least somewhat interpretable. The move toward ML-based systems brought a real leap in predictive power, but it also made the decision process much harder to audit. Rokach (2010) pointed out that ensemble methods like Random Forests, though highly effective, offer limited transparency about which features actually drive outcomes [6].

The fairness literature initially focused on post-hoc fixes—adjusting the decision threshold differently for each demo-

graphic group after training. Hardt, Price, and Srebro (2016) formalised this as “equality of opportunity,” requiring that qualified candidates from all groups have equal chances of being selected [7]. The problem is that threshold adjustment does not touch the model’s internals. If the training data is skewed, the model’s learned representations remain skewed, and the fix is always fighting upstream.

This led researchers toward pre-processing: changing the training data before the model ever learns from it. Kamiran and Calders (2012) introduced re-weighting, where under-represented groups receive higher sample weights to counteract the effects of historical bias [8]. Zafar et al. (2017) went a step further, embedding fairness constraints directly into the learning objective so the model simultaneously optimises for accuracy and demographic parity [9].

What stands out from surveying this work is that most research treats fairness as a one-time correction applied at a single point in the pipeline. No existing study uses the EEOC four-fifths rule as a live, iterative trigger embedded inside a Random Forest hiring pipeline. That is the gap this paper addresses.

III. METHODS AND METHODOLOGY

The framework was built around a Random Forest classifier [10], selected for its strong out-of-the-box performance and widespread use in commercial ATS platforms. The system operates as a loop: train, measure bias, and if the DI score is still below the threshold, rebalance the data and train again. Figure 2 illustrates the full pipeline.

Raw Data → Feature Eng. → DI Calc. → DI < 0.80? → Re-Weigh → FairRF → Output

Fig. 2. End-to-End Fair-ATS Pipeline. DI is evaluated each cycle; re-weighting triggers automatically when DI < 0.80 until compliance.

A. Dataset Description

We used a dataset of 10,847 historical job applications drawn from a public HR analytics repository. Each record contains 18 features: years of experience, education level, resume gap length (in months), university tier (rated 1–5), skill match score, and the final hiring decision. Gender served as the protected attribute, with 61.3% of applications from male candidates and 38.7% from female candidates. The data was split 80/20 for training and testing, with group ratios preserved in both sets. A Pearson correlation check was run upfront to identify and remove proxy features before any model saw the data.

B. Fairness Standard: The Four-Fifths Rule

To determine whether a model is being unfair, we used the EEOC’s four-fifths rule [4]. The rule says that if the minority group’s selection rate falls below 80% of the majority group’s selection rate, there is evidence of adverse impact. The Disparate Impact ratio is:

$$DI = SR_{minority} / SR_{majority} \quad (1)$$

where SR denotes the selection rate for each group. Any DI value below 0.80 means the model is failing the four-fifths rule, and re-weighting kicks in automatically. In our baseline data, the minority group had a 31% selection rate against 50% for the majority, giving a DI of 0.62—well short of the legal threshold.

MAJORITY SR = 50% (50/100 hired)	MINORITY SR = 31% (31/100 hired)
DI = 31%/50% = 0.62 < 0.80 THRESHOLD → Re-Weighing Triggered	

Fig. 3. Four-Fifths Rule Applied. Majority SR = 50%, Minority SR = 31%, DI = 0.62 < 0.80, mandating re-weighting. After Fair-RF: DI = 0.84.

C. Bias Mitigation: Re-Weighting Algorithm

When the DI check fails, re-weighting is applied to the training data [8]. The idea is conceptually simple: give more weight to records where a minority candidate was hired, and less weight to records where the outcome looks shaped by historical bias rather than genuine merit. The weight assigned to each training record is:

$$W(G, Y) = [P(G) \times P(Y)] / P(G, Y) \quad (2)$$

where G is the candidate’s demographic group and Y is the hiring outcome. When $W(G, Y) > 1.0$, that record is amplified during training—the model pays more attention to minority candidates who were genuinely hired. Over multiple iterations, this gradually teaches the model to recognise real potential rather than just replay historical patterns.

D. Feature Engineering and Proxy Elimination

Beyond re-weighting, we cleaned up the feature set before training. Any feature with a Pearson correlation above $|r| = 0.40$ with gender was removed. In practice, this meant dropping postal code region, university prestige tier, and resume gap duration—all of which showed statistically significant correlations with gender in our dataset ($p < 0.01$). Leaving these in would have left backdoor channels through which demographic information could still shape predictions, even after re-weighting [2].

IV. EXPERIMENTAL RESULTS

We compared two models on the held-out test set of 2,170 applications: the original uncorrected Random Forest and the proposed Fair-RF after iterative re-weighting. Three metrics were tracked: Accuracy, F1-Score, and DI Score.

TABLE I
MODEL EVALUATION SUMMARY

Model	Accuracy	F1-Score	DI Score
Uncorrected RF (Baseline)	92.1%	0.89	0.62
Fair-RF (Proposed)	90.4%	0.88	0.84

The DI score climbed from 0.62 to 0.84 across five re-weighting cycles, clearing the legal threshold. Accuracy dropped by 1.7%, but this number deserves a closer reading: it is better

understood as bias-removal noise than as genuine performance degradation. The original model’s high accuracy was partly a product of correctly predicting that minority candidates would be rejected—because that is exactly what the biased training data showed. When re-weighting begins amplifying successful minority candidates, the model starts selecting people it previously rejected. Against biased ground-truth labels, that registers as an error. It is not.

The F1-Scores tell the cleaner story: 0.89 versus 0.88 is essentially the same number. Precision and recall are both preserved. The model is just as good at identifying genuinely qualified candidates; it has simply stopped penalising them for belonging to the wrong demographic group.

Metric	Uncorrected RF	Fair-RF
Accuracy	92.1%	90.4% ($\Delta = -1.7\%$)
DI Score	0.62 (NON-COMPLIANT)	0.84 (COMPLIANT, $\Delta = +0.22$)
F1-Score	0.89	0.88 ($\Delta = -0.01$)

Fig. 4. Performance Comparison. Accuracy and F1-Score remain near-identical; DI Score improves from 0.62 to 0.84, crossing the EEOC compliance threshold.

Metric	Iter 1	Iter 2	Iter 3	Iter 4	Iter 5
DI Score	0.62	0.71	0.77	0.81	0.84
Accuracy	91.9%	91.4%	90.9%	90.5%	90.4%

Fig. 5. DI Score and Accuracy Convergence Across Re-Weighting Iterations. DI rises steadily from 0.62 to 0.84 over five cycles; accuracy loss remains minimal throughout.

V. DISCUSSION

The results carry a nuance worth sitting with. When a model trained on biased data achieves high accuracy, that accuracy is partly a measure of how well the model has learned to replicate the bias—not how well it finds genuinely qualified people. The near-identical F1-Scores (0.89 and 0.88) across both models provide clean empirical support for this interpretation: the real-world predictive value of the model is essentially unchanged; what changed is its willingness to apply the same standards to everyone.

Pre-processing was chosen over in-processing or post-processing fairness interventions for a practical reason: it is classifier-agnostic. Any downstream model benefits from a corrected training distribution without requiring changes to its architecture or training objective. An HR team running a vendor-supplied model can apply the re-weighting step to their next data refresh without touching the model itself.

The iterative loop design matters too. A single-pass correction risks under- or over-correcting depending on the initial data distribution. By re-evaluating the DI score after each cycle and triggering re-weighting only when needed, the framework converges on compliance gradually and measurably. Every iteration’s DI score and accuracy are logged, creating an audit trail that regulators or internal compliance teams can actually inspect.

There are real limitations to acknowledge. The framework currently treats gender as binary and evaluates fairness along

a single demographic axis at a time. Real hiring data involves intersectional identities—a candidate may belong to multiple protected groups at once, and correcting for one axis can inadvertently amplify disparities along another. Extending the DI calculation to handle multi-attribute intersectionality is the most pressing item on the future work agenda.

VI. CONCLUSION

This paper set out to show that a hiring model can be made legally fair without gutting its accuracy. The evidence supports that claim. By using the EEOC four-fifths rule as an iterative trigger and re-weighting as the correction mechanism, the Fair-RF model achieved a DI score of 0.84—comfortably above the 0.80 compliance threshold—while losing only 1.7% in accuracy. The near-identical F1-Scores confirm the model has not lost its ability to find good candidates; it has just lost its habit of discriminating against certain groups of them.

The practical upside is that HR teams can adopt this framework without tearing down what they already have. Because the re-weighting step sits upstream of the classifier, it can be inserted in front of whatever model a company already runs. The DI auditor checks for compliance after each training cycle and flags when intervention is needed—the system corrects itself rather than waiting for a manual review.

The most immediate next step is adding SHAP values [11] so candidates can see which features affected their score—something that will likely be required under the EU AI Act’s transparency provisions. Beyond that, the framework could be extended to cover intersectional fairness and tested in other high-stakes domains like university admissions, healthcare resource allocation, and credit decisions.

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